

TalentInsight – Complete Setup & Working Guide

1. Objective

This document explains how to successfully implement Resume Upload, Job Matching, Skill Gap Analysis, and Course Recommendation features.

2. Requirements

- Python 3.10+
- FastAPI
- MySQL
- React (Vite)
- RapidAPI Key

3. Resume Upload Flow

User logs in → Dashboard → Upload Resume → Backend extracts skills → System decides:

- If job match $\geq 25\%$ → Show Jobs
- If no match → Show Skill Gap + Courses

4. Job Matching Logic

- Fetch jobs using JSearch RapidAPI
- Compare resume skills with job description
- If match percentage $\geq 25\%$, job is returned

5. Skill Gap Logic

If no jobs matched:

- Missing skills are identified
- Courses are recommended for each skill
- Fallback courses are always returned even if API fails

6. Guaranteed Course Display

System forces fallback courses when API returns empty to avoid blank UI.

7. Clicking Course

Each course has a URL → Clicking opens actual learning page in browser.

8. RapidAPI Setup

- Go to <https://rapidapi.com>
- Subscribe to:
 - a) JSearch API (for jobs)
 - b) Udemy Paid Courses API (for courses)
- Copy API Key into `job_api.py` and `course_api.py`

9. Folder Structure

backend/

main.py

resume_parser.py

job_api.py

course_api.py

models.py

frontend/

pages/Dashboard.jsx

pages/JobMatching.jsx

pages/SkillGap.jsx

10. Common Fixes

- If no courses show → check API key
- If PDF not read → install PyPDF2
- If CORS error → allow localhost:5173

11. Final Result

User uploads resume → gets jobs or skill improvement courses → clicks course → starts learning.

You now have a complete, working TalentInsight pipeline.